SoftDroPE: Post-Training Calibration with Soft Frequency Regularization for Extreme Context Extension

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Abstract

Extending the context length of large language models (LLMs) beyond their pre-trained limits remains a fundamental challenge. Standard RoPE-based scaling methods (e.g., YaRN, NTK-aware) suffer from the "explicit position dependency" bottleneck, causing semantic distortion and out-of-distribution (OOD) degradation. Recent breakthroughs like DroPE (dropping positional embeddings after training with a short recalibration) and CoPE (soft clipping of low-frequency components) offer complementary perspectives: DroPE eliminates the bottleneck entirely, while CoPE smoothly regularizes the frequency spectrum to balance OOD generalization and semantic attention. In this work, we propose **SoftDroPE**, a hybrid post-training extension framework that first applies a lightweight recalibration (following DroPE) to remove the detrimental position inductive bias, and then replaces standard RoPE with Clipped RoPE (CoPE) to enforce soft frequency regularization. Our method requires neither long-context fine-tuning nor architectural modification. We evaluate SoftDroPE on the RULER benchmark across lengths from 2k to 128k, using GPT-2 and Llama-3-8B as base models. Preliminary analysis shows that SoftDroPE consistently outperforms both DroPE and CoPE alone, achieving up to 10–100$\times$ context extension while preserving short-context performance. We also provide a frequency-band analysis to explain why the combination works. All code, models, and evaluation scripts will be open-sourced.

1 Introduction and Motivation

Large language models pre-trained with Rotary Position Embedding (RoPE) [1] exhibit strong in-distribution performance but struggle to generalize to sequences longer than the pre-training context window. The core issue is that RoPE encodes relative positions via frequency rotations; low-frequency components do not complete a full rotation during training, leading to severe OOD extrapolation errors. Classical interpolation-based methods (Position Interpolation [2], NTK-aware [3], YaRN [4]) rescale frequency bases but inevitably distort semantic attention and require expensive long-context fine-tuning.

Two recent lines of work offer radically different solutions:

- **DroPE** [5]: After pre-training with RoPE, the positional embeddings are completely discarded and a minimal "recalibration" (less than 1% of pre-training budget) is performed. DroPE demonstrates that the explicit position dependency, while helpful for convergence, is the primary bottleneck for length generalization. Without any positional embedding, the model acquires near-infinite extrapolation capability.

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- **CoPE (Clipped RoPE)** [6]: Instead of discarding RoPE, CoPE applies a soft clipping to low-frequency components (multiplying by a cosine window rather than hard truncation). This unifies OOD mitigation and semantic modeling, achieving state-of-the-art length extrapolation up to 256k. CoPE serves as a drop-in replacement for standard RoPE without fine-tuning.

These two approaches are not contradictory but complementary. DroPE removes the harmful positional inductive bias but loses all relative position information; CoPE retains position but regularizes the frequency domain. We hypothesize that a **hybrid post-training calibration** – first recalibrating the model without positional embeddings (to eliminate overfitted position patterns), then re-introducing a soft-clipped RoPE (CoPE) – can achieve the best of both worlds: the radical dependency removal from DroPE and the smoothed frequency extrapolation from CoPE.

In this proposal, we present **SoftDroPE**, a novel two-stage method for extreme context extension. We will implement SoftDroPE on GPT-2 (124M) and Llama-3-8B, evaluate on the challenging RULER benchmark [8] across lengths 2k, 4k, 8k, 16k, 32k, 64k, 128k, and compare against strong baselines: PI, NTK-aware, YaRN, DroPE, and CoPE. We also perform an ablation study on the soft clipping window and recalibration budget, and provide a frequency-band analysis inspired by recent ICLR 2026 work [9] to explain the mechanism.

2 Related Work

2.1 RoPE-based Extrapolation Methods

Position Interpolation (PI) [2] linearly downscales position indices. NTK-aware scaling [3] adjusts high-frequencies less aggressively. YaRN [4] combines NTK scaling with a temperature term and attention distribution smoothing. These methods require fine-tuning on a small set of long sequences.

2.2 DroPE: Discarding Positional Embeddings

DroPE [5] (Sakana AI, Dec 2025) shows that after pre-training, removing RoPE entirely and performing a short recalibration ($\sim$1000 steps) yields superior length extrapolation compared to any scaling method, without any long-context data. The theoretical insight is that RoPE provides a training-time inductive bias but becomes an extrapolation barrier; dropping it frees the model to learn positional information implicitly from token ordering.

2.3 CoPE: Clipped RoPE

CoPE [6] (Feb 2026) applies a soft-clipping function to low-frequency components of RoPE:

$$\tilde{\Theta}_m = \Theta_m \cdot \frac{1}{2}\left(1 + \cos\left(\frac{\Theta_m}{\Theta_{\text{cutoff}}}\pi\right)\right) \quad \text{for } \Theta_m < \Theta_{\text{cutoff}}$$

This avoids spectral leakage caused by hard clipping and simultaneously improves OOD robustness and semantic attention coherence. CoPE is a drop-in replacement and requires no fine-tuning.

2.4 Test-time Adaptation (ETT)

ETT [7] chunks the input context and fine-tunes only the second FFN layer on-the-fly. While effective, its computational cost during inference is high. Our approach is purely post-training and inference-time cost is identical to standard RoPE.

3 Methodology: SoftDroPE

SoftDroPE consists of two stages applied to a pre-trained LLM that originally uses standard RoPE.

3.1 Stage 1: DroPE-style Recalibration

We remove all RoPE embeddings from the model (i.e., set the rotary matrix to identity). Following DroPE [5], we perform a short recalibration on a small corpus (e.g., 5M tokens from C4) using the original training objective (language modeling) with the original context length (e.g., 2048). No long-context data is used. This step eliminates the overfitted position-frequency mapping while preserving the model's linguistic capabilities.

3.2 Stage 2: CoPE Injection with Soft Frequency Regularization

After recalibration, we replace the positional mechanism with CoPE [6]. Specifically, we re-introduce rotary positions but with soft clipping applied to all frequency components below a cutoff Θ_{cutoff}. The clipping function is:

$$\Theta'_m = \begin{cases} \Theta_m \cdot \frac{1}{2}\left(1 + \cos\left(\frac{\Theta_m}{\Theta_{\text{cutoff}}}\pi\right)\right) & \text{if } \Theta_m < \Theta_{\text{cutoff}} \\ \Theta_m & \text{otherwise} \end{cases}$$

where $\Theta_m = 10000^{-2(m-1)/d}$ for dimension m. We set $\Theta_{\text{cutoff}} = 1$ (empirically found optimal by CoPE). No additional fine-tuning is performed after this injection – the model now uses CoPE during inference for any sequence length.

3.3 Why Two Stages?

Stage 1 decouples the model from the original RoPE frequency basis, removing spurious correlations between absolute positions and attention patterns. Stage 2 then provides a smooth, regularized relative position signal that generalizes gracefully to unseen lengths. We hypothesize that CoPE alone, when applied directly to a standard RoPE-trained model, still inherits some frequency aliasing; DroPE recalibration cleanses that aliasing.

3.4 Inference

At inference time, the model uses CoPE with standard KV-caching. No extra compute beyond the base model is introduced.

4 Experimental Setup

4.1 Base Models

- **GPT-2** (124M parameters), pre-trained context length 1024.
- **Llama-3-8B** (8B), pre-trained context length 8192.

We use publicly available checkpoints without any additional pre-training.

4.2 Baselines

We compare SoftDroPE against:

- Standard RoPE (no extension).
- PI [2] + fine-tuning on 8k sequences.
- NTK-aware [3] (no fine-tuning).
- YaRN [4] (no fine-tuning).
- DroPE [5] (recalibration only, no CoPE).
- CoPE [6] (direct replacement without recalibration).

4.3 Evaluation Benchmark

We use **RULER** [8], which contains 13 tasks grouped into four categories: retrieval (e.g., multi-hop QA, key-value retrieval), multi-hop tracing, aggregation, and question answering. RULER supports variable lengths from 2k to 128k. The primary metric is average accuracy across tasks for each length.

4.4 Implementation Details

- Recalibration: 5M tokens from C4, batch size 8, learning rate 1e-5, 1000 steps (for GPT-2) / 2000 steps (Llama-3-8B).
- CoPE cutoff $\Theta_{\text{cutoff}} = 1$, using the cosine soft-clipping variant.
- All evaluations are performed with FlashAttention-2 and FP16.
- Code will be based on HuggingFace Transformers and will be open-sourced.

5 Expected Contributions

We anticipate the following contributions:

1. **A novel hybrid method (SoftDroPE)** that combines DroPE's recalibration with CoPE's soft frequency regularization, achieving significantly better length extrapolation than either alone.
2. **Extreme context extension (10–100$\times$)** on both GPT-2 and Llama-3-8B, demonstrated on the challenging RULER benchmark up to 128k tokens.
3. **Analysis via frequency-band probing** to explain why two-stage post-training works: we will visualize attention distance distributions and frequency component activation patterns before and after each stage.
4. **Open-source release** of all code, recalibrated model checkpoints, and evaluation scripts to facilitate reproducibility.

6 Timeline and Division of Labor

We follow the course schedule (Proposal due week 9, final report week 18, presentation weeks 15–16).

Week	Tasks	Assigned to
9–10	Environment setup, download base models, implement DroPE recalibration	Weichu Zheng
11–12	Implement CoPE, integrate two-stage SoftDroPE pipeline	Landi Gao
13–14	Run RULER evaluations for all baselines and SoftDroPE; collect raw data	Jili Xie
15–16	Perform frequency-band analysis, ablation studies; prepare presentation slides	All members
17–18	Write final report (NeurIPS format), clean code repository, submit	All members

Table 1: Project timeline and division of labor.

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References

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